

SETH AYUMU
(Meng, Petroleum Engineering, BSc. Mechanical Engineering)

Edmonton, AB, Canada

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🌐 <https://mrayumu.github.io>

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EDUCATION

University Of Alberta, Edmonton AB

September 2023 – April 2025

- Master of Engineering Degree, Petroleum Engineering
- **Projects:**
 - **A Data Mining Approach to Assess Field Scale CO2 EOR and Sequestration Performance Correlated to Geological and Reservoir Characteristics:** Pioneering framework that enhances strategic decision-making for infill development. Presented at the SPE Canadian Energy Technology Conference and Exhibition 2025 and still undergoing peer review for publication in the SPE Journal
 - **Supply Chain Optimization for DAAB Inc.:** Developed a cost-optimization model using LINDO, resulting in a more efficient supply chain network with potential cost savings of up to 15%
 - **Reservoir Characterization of The Volve Field:** Analyzed production data to characterize the subsurface reservoir, providing valuable insights for improving resource extraction strategies
 - **Gas Lift System Design:** Used PIPESIM to simulate a well and design a gas lift system, significantly enhancing production efficiency and optimizing well performance.
 - **Multiphase Flow Simulation:** Implemented the Drift Flux Model using MATLAB to accurately simulate multiphase flow through production pipes, enhancing the understanding of fluid dynamics in petroleum production
 - **Reservoir Simulator Development:** Created a custom reservoir simulator using MATLAB to model fluid flow phenomena through porous media, contributing to advanced predictive capabilities in reservoir engineering
- Member of Interdisciplinary Consulting Association

University For Development Studies, Tamale, Ghana

September 2017–July 2021

- Bachelor of Science Degree, Mechanical Engineering (4.3 CGPA)
- Led an educational outreach program to rural communities, demonstrating adaptability, problem-solving, and impactful communication by directly benefiting over 1,000 individuals
- Managed budget and strategic funding allocation as Finance Committee Chair of the Student Representative Council, showcasing leadership, analytical thinking, and effective team coordination

WORK EXPERIENCE

Data Management Specialist — Freelance

Akuapem North Municipal Assembly, Akropong— Ghana

September 2022–August 2023

- Developed an innovative building permit applications data management system, applying critical problem-solving skills to enhance service speed by 10× and reduce paper use by 80%.
- Improved record-keeping accuracy by over 90%, ensuring data integrity and operational efficiency.

Customer Service Associate & Administrative Assistant (Ghana National Service Personnel)

Akuapem North Municipal Assembly, Akropong— Ghana

October 2021 –August 2022

- Streamlined administrative workflows by initiating a user-friendly database system to reduce errors and processing time
- Delivered efficient customer service, improving client satisfaction through proactive communication and team collaboration

CERTIFICATIONS

- PMP® Certification Training (December 2023), Gold Standard Certifications
- Diploma in Supply Chain Management (March 2023), Alison

PROFESSIONAL DEVELOPMENT

Aspire Institute, Virtual

March 2023–October 2023

- Selective academic and professional development program for first-generation college students with applicants from over 100 countries
- Coursework completed includes a virtual leadership course, live interdisciplinary academic sessions and professional development workshops with Harvard Business School faculty members
- Developed a growth mindset and leadership abilities while collaborating in a global, multidisciplinary environment
- Leveraged skills in problem-solving and effective communication to gain insights from members of the Harvard Business School Faculty during live case study sessions

SKILLS

- Web Application Development in Python (Front-end & Back-end)
- Data Analysis and Visualization
- Artificial Lift Design
- Nodal Analysis of Oil and Gas Systems
- Well Log Interpretation and Formation Evaluation
- Well Test Analysis
- Excel VBA Programming
- Reservoir Simulation
- LINDO

COMMUNITY/PERSONAL

- Volunteer with the Canadian Red Cross at the HELP Depot
- Enjoy reading non-fiction. *Outliers* remains my most impactful read